

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is -0.5 . For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

Perceptron Classification (Pass/Fail)

We're given:

- Inputs:
 - Study hours = 2
 - Attendance = 1
- Weights:
 - $w_1=0.6$, $w_2=0.4$
- Bias: $b=-0.5$

Step 1: Compute Net Input

$$\begin{aligned}\text{Net input} &= (w_1 \cdot x_1) + (w_2 \cdot x_2) + b \\ &= (0.6 \cdot 2) + (0.4 \cdot 1) - 0.5 \\ &= 1.2 + 0.4 - 0.5 = 1.1\end{aligned}$$

Step 2: Apply Step Activation

Step function:

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Since $1.1 \geq 0$, output = **1**.

 **Conclusion:** The perceptron predicts **Pass**.

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

We're given:

- Initial weight: $w=0.5$
- Learning rate: $\eta=0.1$
- Input: $x=2$
- Actual output: $y=0.6$
- Target output: $t=1$

Step 1: Compute Error

$$\text{Error} = t - y = 1 - 0.6 = 0.4$$

Step 2: Weight Update Rule

Delta rule:

$$\begin{aligned}\Delta w &= \eta \cdot \text{Error} \cdot x \\ &= 0.1 \cdot 0.4 \cdot 2 = 0.08\end{aligned}$$

Step 3: New Weight

$$w_{\text{new}} = w + \Delta w = 0.5 + 0.08 = 0.58$$

✅ **Conclusion:** After one iteration, the updated weight is **0.58**.

✨ Final Answers

1. Perceptron output = 1 → Student Passes
2. Updated weight after training = 0.58